

CSE427 | Machine Learning

Name:	ID:	Sec:
-------	-----	------

1. a) Given, $y = \frac{1}{1+e^{-x}}$, show that $\frac{dy}{dx} = y(1 - y)$ [3]

b) The below neural networks hidden layer uses sigmoid function and output layer uses linear function, show that

$$w_{24(new)} = w_{24(old)} - \eta \cdot (w_{47}\delta_7 + w_{48}\delta_8 + w_{49}\delta_9) \cdot y_4(1 - y_4) \cdot x_2$$

Where, $\delta_k = \frac{dL}{dy_k}$. The loss function is given below. [10]

$$Loss = \frac{1}{2} \{ (Ob_7 - y_7)^2 + (Ob_8 - y_8)^2 + (Ob_9 - y_9)^2 \}$$

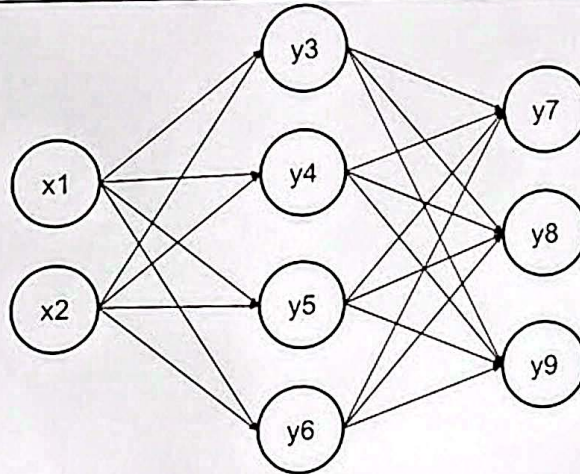


Fig.1. Neural Network

c) Given,
 $x_1 = 1.5, x_2 = 2.5, y_7 = 1, y_8 = 1, y_9 = 0, \eta = 0.1, w_{all} = 1.5, b_{hidden} = 3.5$ and $b_{output} = 2.0$
 Find updated weight of w_{24} . [2]